CNNP-SKILL-2

THREE-POINT PACKET CAPTURE AND NETWORK PERFORMANCE ANALYSIS

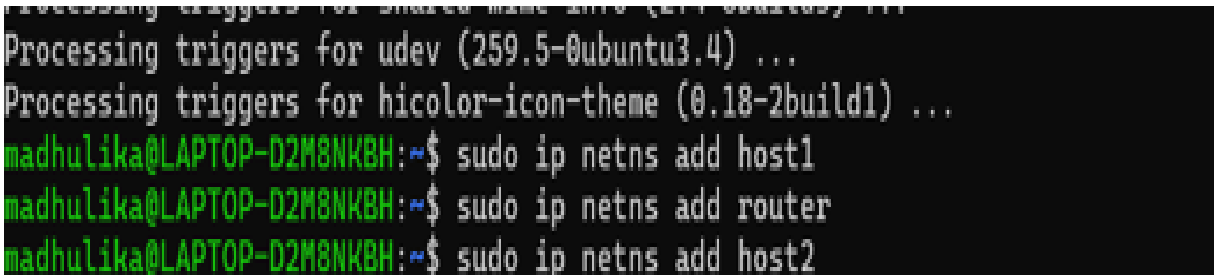
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Roll No: 2520090056

To establish a virtual network consisting of Host1, a Router, and Host2, configure IP connectivity and network conditions, capture packets at three different locations, and examine packet details, message transmission, network delay, packet loss, and TCP communication using Linux commands and Wireshark.

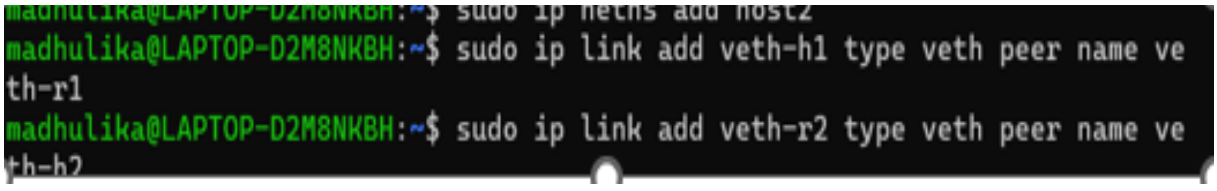
Step 1 – Setting Up Network Namespaces

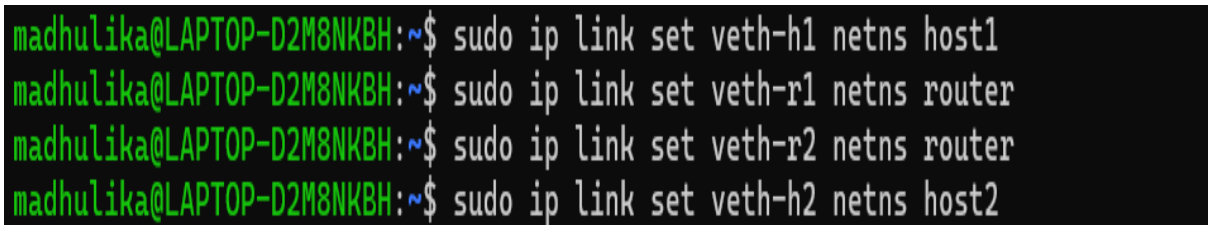
Three separate network namespaces named host1, router, and host2 were created in Ubuntu. These namespaces act as independent network devices, where Host1 works as the sender, Host2 acts as the receiver, and the router connects both hosts to enable communication.



Step 2 – Creating Virtual Ethernet Connections

Virtual Ethernet (veth) interfaces were created to establish connections between Host1 and the router, and between the router and Host2. Each veth pair consists of two connected endpoints that act like a virtual network cable, allowing data to travel between the network namespaces.





Step 3 – Configure Host1 and its interface

Moved the Host1-side interface into the host1 namespace, assigned its IP address and brought the interface up.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/3
0 dev veth-h1
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 ip link set veth-h1 up
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 ip link set lo up
```

Step 4 – Configuring the Router

Configured the router interfaces with IP addresses and enabled IPv4 forwarding to allow communication between Host1 and Host2.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router ip addr add 10.0.1.2/
30 dev veth-r1
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router ip addr add 10.0.2.1/
30 dev veth-r2
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router ip link set veth-r1 u
p
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router ip link set veth-r2 u
p
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router ip link set lo up
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 5 – Configuring Host2

The Host2-side interface was moved into the host2 namespace, assigned the IP address and activated to establish network connectivity.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/3
0 dev veth-h2
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host2 ip link set veth-h2 up
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host2 ip link set lo up
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 6 – Configuring Routing Information

Added routing entries to Host1 and Host2 so that packets destined for the opposite network are forwarded through the router, enabling communication between both hosts.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router sysctl -w net.ipv4.ip
_forward=1
net.ipv4.ip_forward = 1
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 ip route add 10.0.2.0/
30 via 10.0.1.2
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host2 ip route add 10.0.1.0/
30 via 10.0.2.1
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 7 – Testing Network Connectivity

Used the ping command from Host1 to Host2 (10.0.2.2) to check whether packets were successfully transmitted and replies were received across the virtual network.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=3.27 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.059 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.059 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.059/1.128/3.267/1.512 ms
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 8 – Configuring Network Delay and Packet Loss

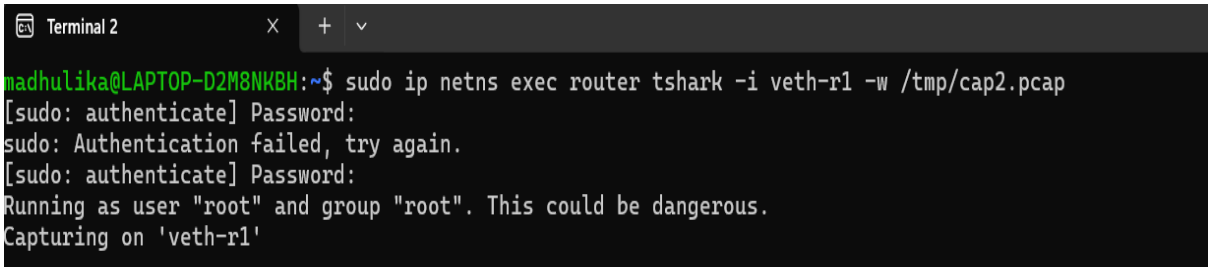
Applied network delay of 20 ms and 1% packet loss using the tc netem command on the router interface to simulate network conditions.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc show dev vet
h-r2
qdisc noqueue 0: root refcnt 2
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc add dev veth
-r2 root netem delay 20ms loss 1%
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc show dev vet
h-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 13725878
156063232424
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 9 – Capturing Packets at Host1

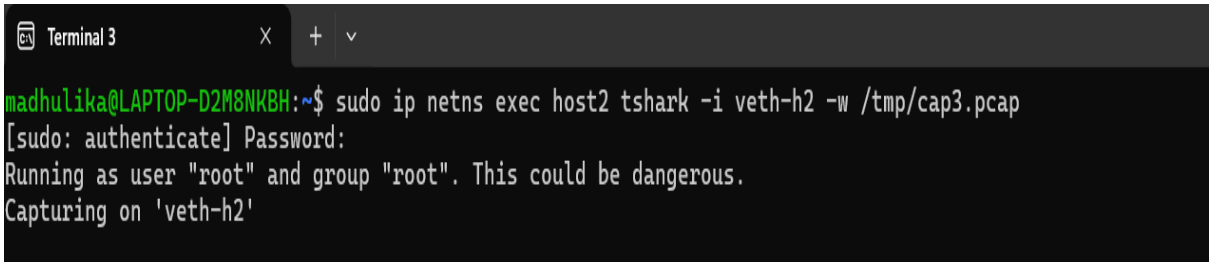
Started packet capturing on Host1's network interface using tshark and saved the captured packets in /tmp/cap1.pcap for further analysis.

```
h-r2 root netem delay 20ms loss 1%
Object "netnus" is unknown, try "ip help".
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc show dev vet
 h-r2 root netem delay 20ms loss 1%
What is "h-r2"? Try "tc qdisc help".
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netus exec router tc qdisc show dev vet
h-r2 root netem delay 20ms loss 1%
Object "netus" is unknown, try "ip help".
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc show dev vet
h-r2
qdisc noqueue 0: root refcnt 2
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc add dev veth
-r2 root netem delay 20ms loss 1%
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec router tc qdisc show dev vet
h-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 13725878
156063232424
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /
tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```



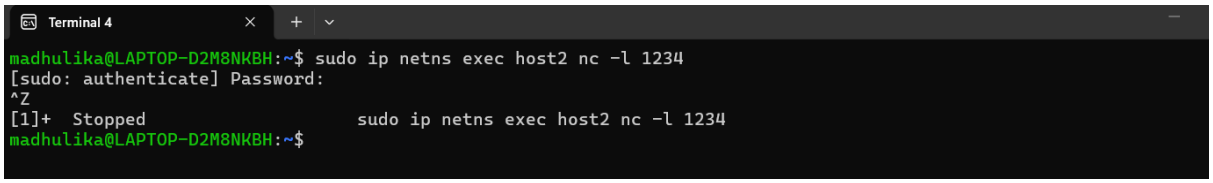
Step 10 – Capturing Packets at the Router

Started packet capturing on the router interface using tshark and saved the captured traffic in /tmp/cap2.pcap for analysis.



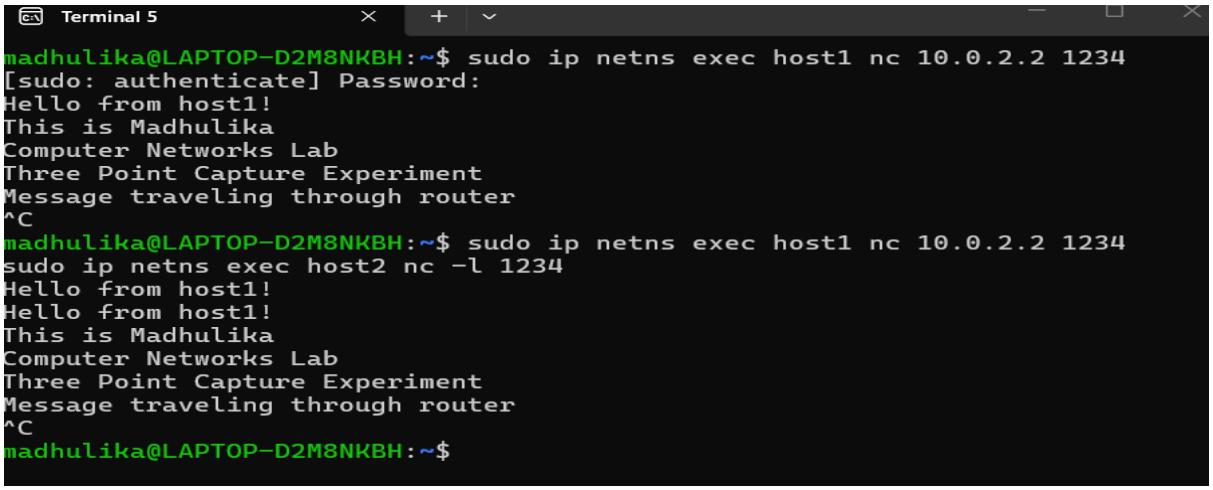
Step 11 – Capturing Packets at Host2

Started packet capturing on Host2's network interface using tshark and saved the captured traffic in /tmp/cap3.pcap. Captures from all three points were used to compare packet transmission across the network.



Step 12 – Sending a TCP Test Message

Used Netcat on TCP port 1234 to send a test message from Host1 to Host2. The messages were received successfully and recorded in the packet captures.



Step 13 – Verifying the Capture Files

Changed the permissions of all three packet capture files and checked their file sizes to confirm that the captured data was saved successfully.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pca
p /tmp/cap3.pcap
madhulika@LAPTOP-D2M8NKBH:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/c
ap3.pcap
-rw-r--r-- 1 root root  460 Nov  7 07:03 /tmp/cap1.pcap
-rw-r--r-- 1 root root 6.7K Nov  7 13:39 /tmp/cap2.pcap
-rw-r--r-- 1 root root 6.3K Nov  7 13:39 /tmp/cap3.pcap
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 14 & 15– Analysing IP TTL Values

Used tshark to extract and compare the TTL values from all three packet captures. The results showed a TTL of 64 at Host1 and the router, and 63 at Host2 due to router forwarding.

```
madhulika@LAPTOP-D2M8NKBH:~$ sudo tshark -r /tmp/cap2.pcap -Y "tcp.port
== 1234 && tcp.len > 0" -T fields -e tcp.payload | tr -d ':\n' | xxd -r
-p
Running as user "root" and group "root". This could be dangerous.
Hello from host1!
This is Madhulika
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

Step 16– Compare Packet Timestamps

Extracted frame.time_epoch timestamps for TCP data packets from cap1.pcap and cap3.pcap. The timestamp difference for the first corresponding packets was approximately 0.02108 seconds (21.08 ms), which is close to the configured 20 ms network delay. This comparison assumes that the capture timestamps are sufficiently aligned.

```
madhulika@LAPTOP-D2M8NKBH:~$ t1=$(tshark -r /tmp/cap1.pcap -Y "tcp.port == 1234 && tcp.len > 0" -T fields -e frame.time_
epoch | head -1)
madhulika@LAPTOP-D2M8NKBH:~$ t3=$(tshark -r /tmp/cap3.pcap -Y "tcp.port == 1234 && tcp.len > 0" -T fields -e frame.time_
epoch | head -1)
madhulika@LAPTOP-D2M8NKBH:~$ awk -v a="$t1" -v b="$t3" 'BEGIN {d=b-a; if(d<0)d=-d; printf "Time difference: %.5f seconds
 (%.2f ms)\n",d,d*1000}'
Time difference: 0.02029 seconds (20.29 ms)
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 17 – Check TCP Retransmission Indications

Examined all three packet capture files using the tcp.analysis.retransmission filter in tshark. No matching packets were found in cap1.pcap, cap2.pcap, or cap3.pcap. This indicates that no TCP retransmission events were detected in the captured traffic, although it does not guarantee that packet loss did not occur.

```
madhulika@LAPTOP-D2M8NKBH:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
madhulika@LAPTOP-D2M8NKBH:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
madhulika@LAPTOP-D2M8NKBH:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
madhulika@LAPTOP-D2M8NKBH:~$
```

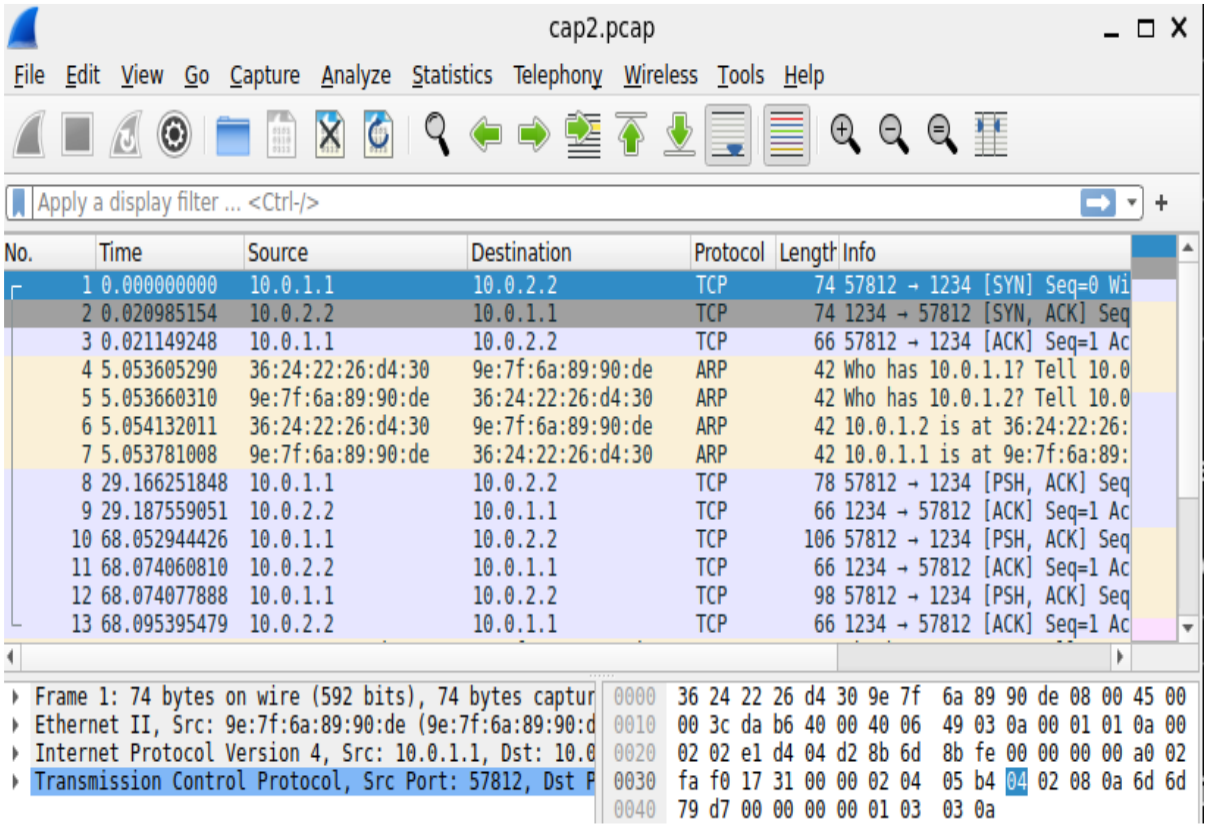
Step 18 – Analyze Network Performance Using MTR

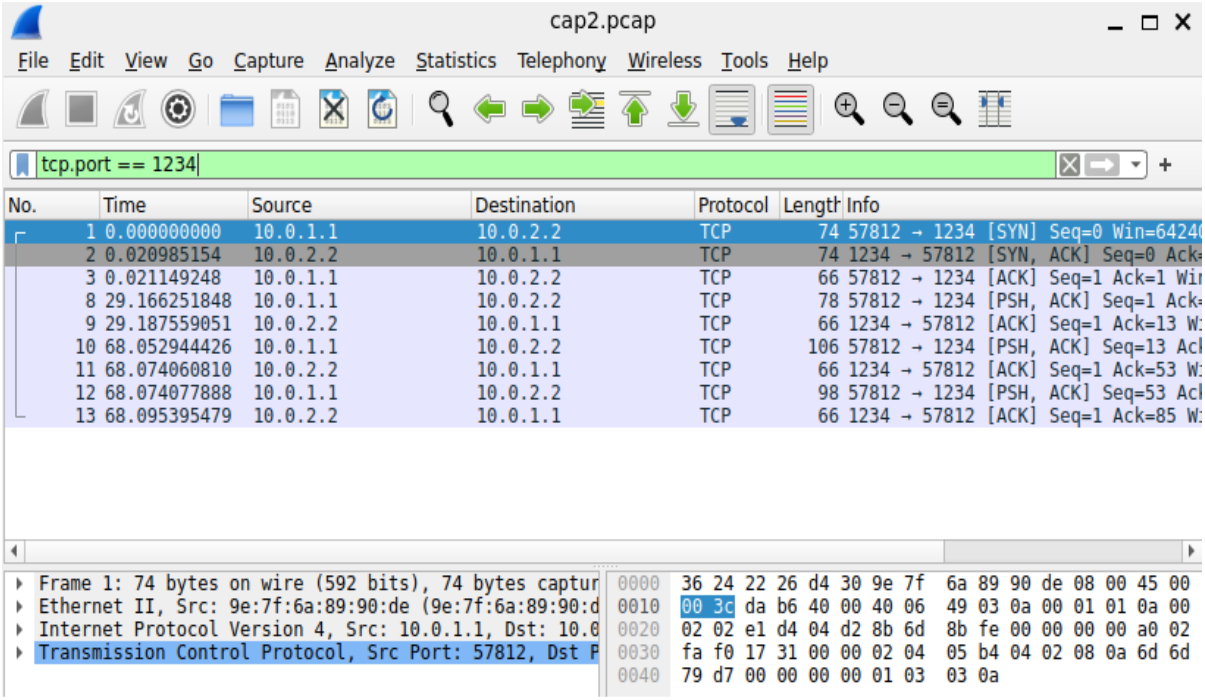
Executed the MTR tool from Host1 to Host2 (10.0.2.2) to evaluate network performance. The results indicated 0.0% packet loss with an average RTT of 0.2 ms at the router (10.0.1.2), while Host2 (10.0.2.2) showed 10.0% packet loss and an average RTT of 21.3 ms. These results were based on 10 probes, so the observed packet loss represents a limited sample.

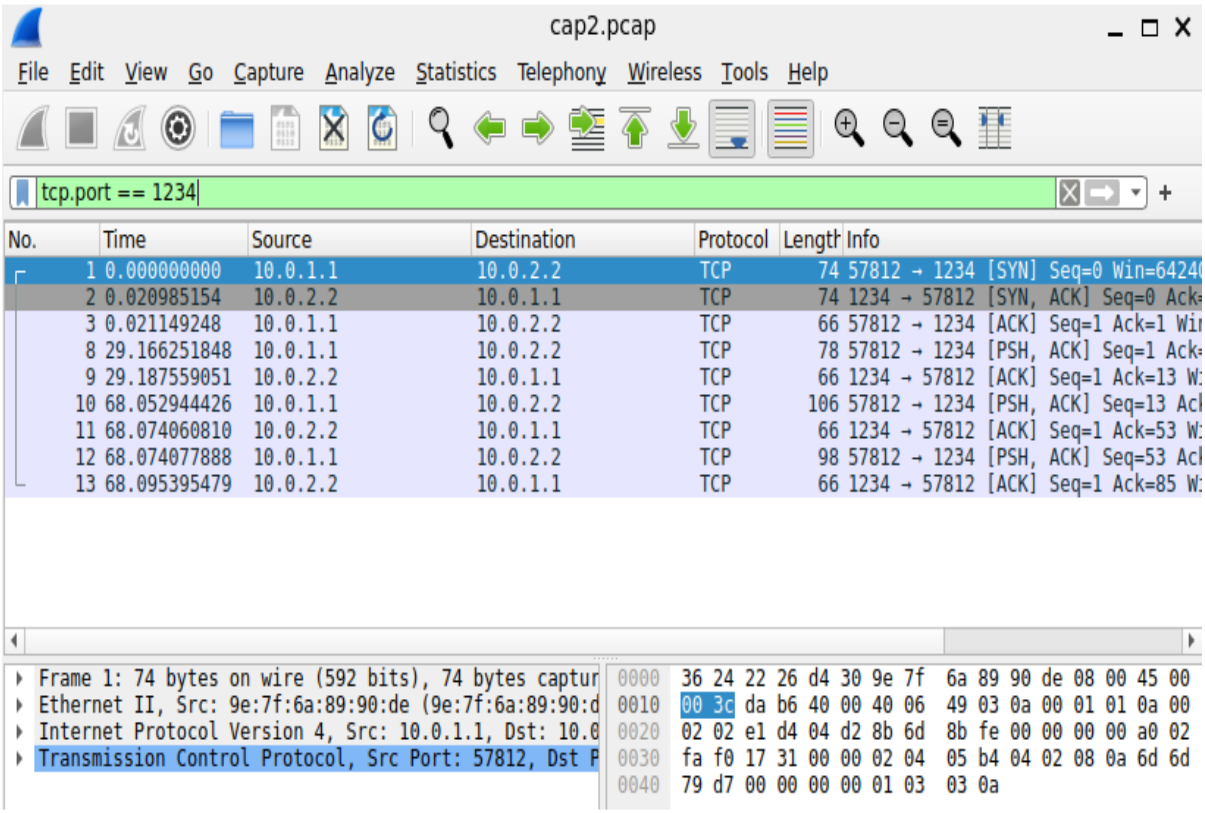
```
madhulika@LAPTOP-D2M8NKBH:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-11T17:19:27+0000
HOST: LAPTOP-D2M8NKBH             Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.2   0.1   0.4   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.6  20.6  20.3  20.9   0.2
madhulika@LAPTOP-D2M8NKBH:~$
```

Step 19 – Inspect packets in Wireshark

Opened the router packet capture file (cap2.pcap) in Wireshark and applied the filter tcp.port == 1234 to view the TCP communication. Examined the selected packets to study the Frame, Ethernet, IPv4, TCP, and Data fields and understand how the messages were transmitted through the network.

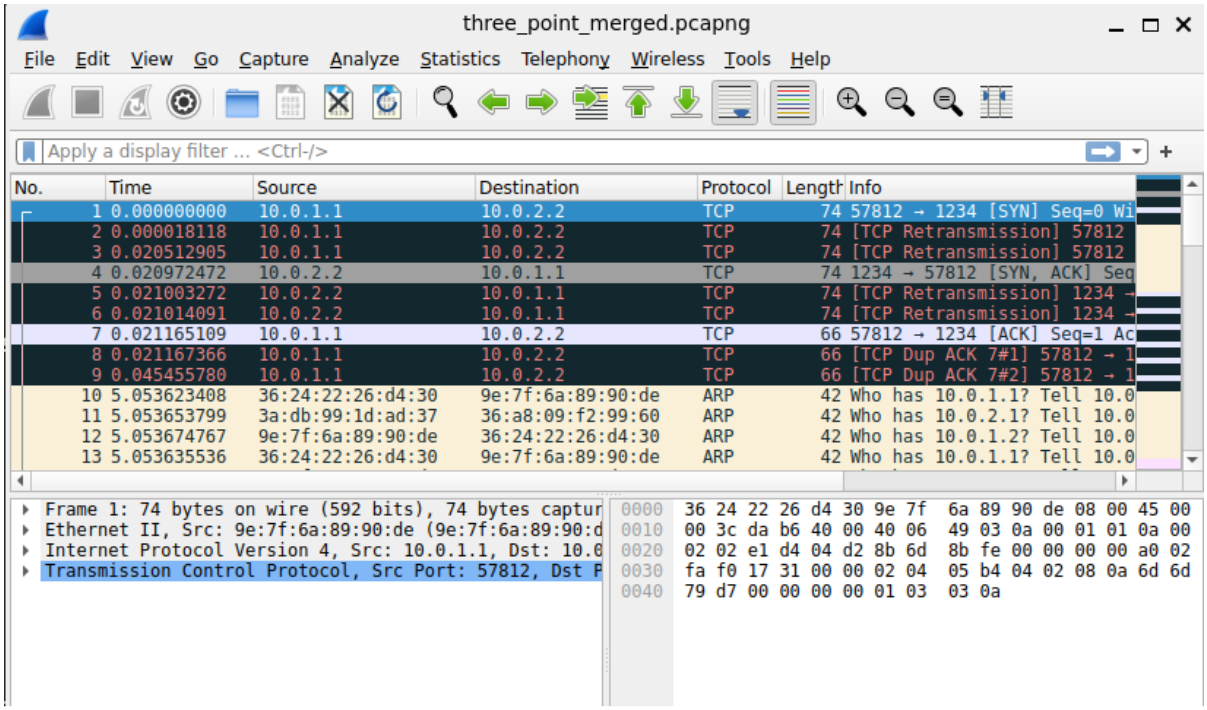






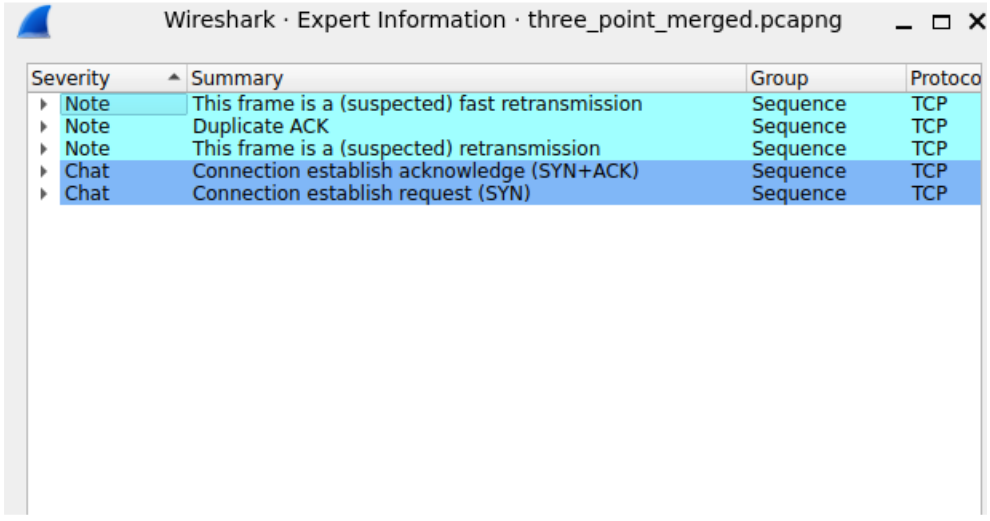
Step 20 – Merge the three packet captures

Combined the three capture files (cap1.pcap, cap2.pcap, and cap3.pcap) into a single file named /tmp/three_point_merged.pcapng using mergecap. Verified the merged file with capinfos, which reported 53 packets collected across three interfaces: 18 from veth-h1, 18 from veth-r1, and 17 from veth-h2.



Step 21 – Review Wireshark Expert Information

Accessed the Expert Information panel through Wireshark's Analyze menu to inspect the merged packet capture. The analysis highlighted events such as suspected fast retransmissions, duplicate acknowledgements, and suspected retransmissions. Since the merged file contains packet observations from multiple network interfaces, duplicate observations may influence these results. Therefore, these indications alone cannot confirm end-to-end TCP retransmissions.



Step 22 – Remove the Network Namespaces

Removed the host1, router, and host2 network namespaces using the ip netns del command. Finally, executed ip netns list to verify the cleanup. The command returned no entries, confirming that all experiment-related namespaces were successfully deleted.